

FCFS Appointment Simulation under a Plausible Two-Class Parameterization

Supplementary comparison with the stress-test baseline

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Abstract

The main analysis note used a deliberate stress-test configuration ($\lambda = 100$ arrivals per day against $S = 32$ slots, demand-to-capacity ratio 3.1:1) to expose how the simulation metrics behave under severe capacity constraints. This supplementary note repeats the core comparison under a more plausible, though still stylized, two-class parameterization: an MRI-like diagnostic class and a behavioral-health follow-up class. The comparison setting has 24 daily arrivals, 20 slots, a 28-day booking horizon, and class-specific behavioral parameters from the model formulation note. Thirty independent seeds are used throughout.

The aggregate output improves sharply when demand pressure is reduced. Average utilization rises to 0.934 and overall served rate rises to 0.789, compared with 0.842 and 0.269 in the stress-test baseline. The more important new result is the emergence of a large class-level access gap: Class 1 served rate is 0.881, while Class 2 served rate is 0.661. Under the current pooled FCFS formulation, this gap is mechanically consistent with asymmetric behavioral parameters rather than explicit class priority. Class 1 has a higher delay tolerance, lower baseline no-show probability, and lower cancellation hazard; Class 2 is disadvantaged along all three stages of the access pipeline. Thus, the stress-test note explains the aggregate mechanics, while this comparison shows that aggregate improvement can still conceal substantial class-level divergence.

1 Parameter Configuration

Tables 1 and 2 compare the two configurations. The first table isolates the system-level load reduction. The second table makes the class asymmetry visible directly: the stress-test baseline is symmetric by construction, while the comparison parameterization makes Class 1 more tolerant of delay and less exposed to cancellation or no-show than Class 2.

Parameter	Stress-test baseline	Comparison parameterization
Total arrivals λ per day	100	24
Class 1 arrivals λ_1 per day	50	14
Class 2 arrivals λ_2 per day	50	10
Slots per day S	32	20
Demand/supply ratio	3.1	1.2
Booking horizon H in days	14	28
Burn-in / cooldown in days	30 / 10	60 / 30

Table 1: System-level configuration comparison. The comparison parameterization reduces load from severe oversubscription to mild oversubscription while using a longer booking horizon.

Parameter	Stress C1	Stress C2	Comparison C1	Comparison C2
Balking threshold in days	9	9	21	14
Low-delay balking probability b^{low}	0.00	0.00	0.02	0.04
High-delay balking probability b^{high}	0.50	0.50	0.55	0.72
No-show threshold in days	6	6	21	14
Low-delay no-show probability ξ^{low}	0.00	0.00	0.01	0.15
High-delay no-show probability ξ^{high}	0.30	0.30	0.31	0.51
Daily cancellation probability ϕ	0.10	0.10	0.01	0.02

Table 2: Class-specific behavioral parameters. The stress-test baseline is symmetric across classes. The comparison parameterization is asymmetric: Class 1 has greater delay tolerance and lower post-booking loss risk than Class 2.

2 Outcome Comparison

Tables 3 and 4 report the main outputs for both configurations, each averaged over 30 independent seeds. The aggregate table shows the effect of lower demand pressure. The horizontal class-level comparison then shows the result that is hidden by aggregate metrics alone.

Metric	Stress-test baseline	Comparison parameterization
Average utilization	0.8418	0.9341
Overall served rate	0.2692	0.7890
Mean offered wait in days	9.30	10.87
Mean accepted wait in days	8.35	10.85
Offered minus accepted wait in days	0.95	0.02
Served share	0.269	0.789
Balked share	0.356	0.033
Canceled share	0.323	0.123
No-show share	0.052	0.056
No-offer share	0.000	0.000
Unresolved share	< 0.001	0.000

Table 3: Aggregate outcomes and arrival outcome shares under both configurations, averaged over 30 independent seeds. The stress-test figures are from the controlled no-balking diagnostic run using the same 30-seed protocol.

Metric	Stress C1	Stress C2	Comparison C1	Comparison C2
Served rate	0.269	0.269	0.881	0.661

Table 4: Class-level served rates. The stress-test baseline is symmetric, while the comparison parameterization produces a Class 1 minus Class 2 served-rate gap of 0.220.

3 What Changes and What Does Not

Aggregate utilization and served rate. Both improve substantially. Utilization rises from 0.842 to 0.934; served rate rises from 0.269 to 0.789. The primary driver is the lower demand pressure combined with much lower cancellation hazards. With $\phi = 0.01$ – 0.02 per day instead of 0.10 per day, far fewer booked slots are vacated before service. For a mean delay near 10.9 days, the probability of avoiding cancellation is approximately $0.99^{10.9} \approx 0.90$ for Class 1 and $0.98^{10.9} \approx 0.80$ for Class 2, compared with $0.9^{8.35} \approx 0.42$ in the stress-test baseline. The served-rate gain is also mechanically consistent with the lower demand-to-capacity ratio: the nominal capacity ceiling is $20/24 = 0.833$, and the simulated served rate (0.789) is close to that bound after cancellations, no-shows, and balking.

Balking collapses; the offered/accepted gap nearly vanishes. In the stress-test baseline, mean offered wait is 9.30 days, near the 9-day balking threshold, which produces a 35.6% balked share and a 0.95-day gap between offered and accepted wait. In the comparison parameterization, mean offered wait is 10.87 days, below the Class 1 threshold of 21 days and below the Class 2 threshold of 14 days. Only 3.3% of arrivals balk. Because few high-delay patients are removed from the accepted sample, offered and accepted wait are nearly identical (0.02 days apart). In this lower-load regime, the selection problem in mean accepted wait is much smaller than in the stress-test baseline.

No-show share is similar; the mechanism is different. No-show shares are similar (5.2% versus 5.6% of arrivals), but for different reasons. In the stress-test baseline, many accepted delays exceed the 6-day no-show threshold, so many booked patients face the high-delay no-show probability $\xi^{\text{high}} = 0.30$.

In the comparison parameterization, the 21-day Class 1 threshold means that most accepted Class 1 bookings face only $\xi_1^{\text{low}} = 0.01$. The aggregate no-show share is instead pulled up by Class 2, whose $\xi_2^{\text{low}} = 0.15$ applies even for short-wait offers. Under the current formulation, this makes Class 2 a persistent source of booked-slot loss even when delays are moderate.

A large class access gap emerges. The most important new finding is the 22-percentage-point class served-rate gap shown in Table 4 and Figure 1: Class 1 served rate is 0.881, while Class 2 served rate is 0.661. This is far larger than the near-zero gap at the symmetric stress-test baseline. The horizontal class-level comparison makes this visible directly.

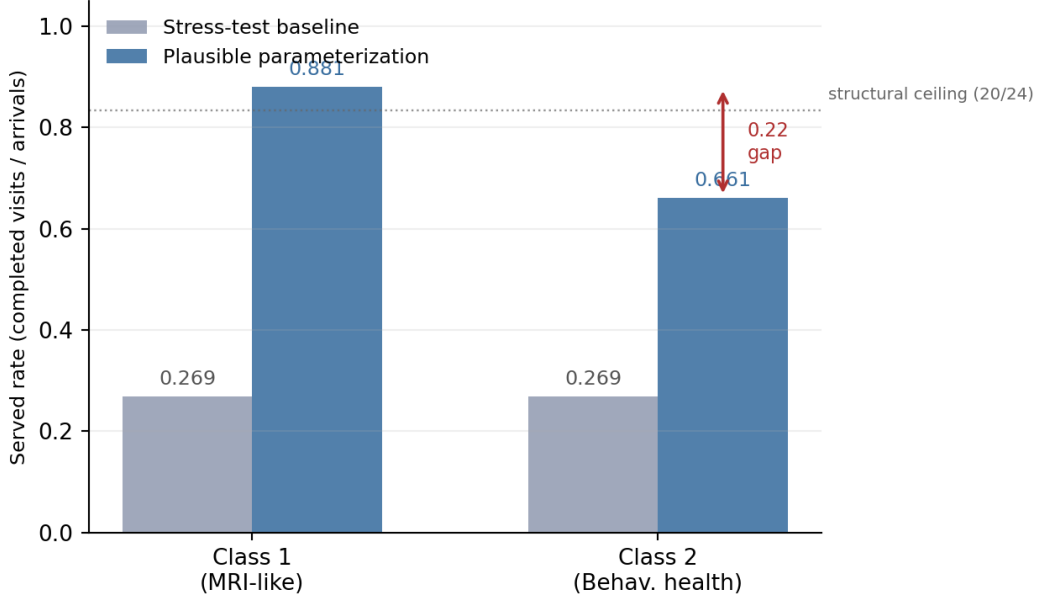


Figure 1: Class-level served rates under the comparison parameterization, with the symmetric stress-test baseline shown for reference. Aggregate served rate improves relative to the stress-test baseline, but access is uneven across classes. Under pooled FCFS, this gap is mechanically consistent with asymmetric behavioral parameters rather than explicit class priority.

The gap is mechanically consistent with three compounding Class 1 advantages: higher delay tolerance (balking threshold 21 rather than 14 days), lower baseline no-show probability (0.01 rather than 0.15), and lower daily cancellation probability (0.01 rather than 0.02). Under pooled FCFS, these advantages act through the shared booking pool. Class 1 accepts more long-delay offers, keeps more booked appointments, and loses fewer visits before service. Class 2 is disadvantaged along the same stages. The output therefore suggests a shared-capacity spillover rather than explicit class priority.

This finding is not recoverable from aggregate metrics alone. Average utilization (0.934) and overall served rate (0.789) both look high. The class gap only appears when class-level served rates are reported separately.

No-offer remains zero; the qualitative horizon dynamics hold. As in the stress-test baseline, no-offer is zero. This does not mean that the system is unconstrained; it means that, under the current formulation, cancellations and daily slot rollover are sufficient to prevent full horizon saturation in the observed runs. This should be read together with the served rate, cancellation share, and class-level served rates rather than interpreted as evidence of unconstrained access.

4 Summary

The stress-test note showed that low served rate can be mechanically consistent with high utilization when demand strongly exceeds capacity. This supplementary comparison shows what changes under a lower-load, asymmetric two-class parameterization. Aggregate outcomes improve: utilization rises to 0.934 and overall served rate rises to 0.789. The loss-channel mix also changes: balking falls sharply, cancellation losses are lower, and no-offer remains zero.

The critical remaining issue is class-level access. Under the current pooled FCFS formulation, Class 1 reaches a served rate of 0.881, while Class 2 reaches only 0.661. This gap is mechanically consistent with asymmetric behavioral parameters: Class 1 is more willing to accept delayed offers, less likely to cancel, and less likely to no-show at low delays. The comparison therefore does not contradict the stress-test note. It extends it: improving aggregate performance does not necessarily remove class-level divergence when patient classes differ in behavioral risk.